

Aurenix Signal

AI Discovery Optimization Platform - Product Roadmap

Signal v1.0 MVP roadmap and Signal v2.0 commercial platform modules. Static PDF converted from the provided React app content.

v1.0 Start 06/15/2026	v1.0 Target Sep 4, 2026	Engines ChatGPT + Gemini
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Signal v1.0 - MVP Roadmap

Prove the hypothesis: targeted content changes increase a brand AI recommendation rate. The Nautikal is the live case study. 12 weeks, 4 phases, working system with a measurable before/after GEO Score delta as the exit criterion.

Phase 0 Foundation & Baseline Weeks 1-2 5 tasks	Phase 1 Prompt Runner (Agent 1) Weeks 3-4 4 tasks	Phase 2 Response Judge & GEO Scoring (Agent 2) Weeks 5-7 6 tasks	Phase 3 GEO Dashboard (MVP UI) Weeks 8-9 4 tasks	Phase 4 GEO Recommender (Agent 3 MVP) + Full Loop Weeks 10-12 5 tasks
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Phase 0 Jun 15 - Jun 26, 2026

Foundation & Baseline

Establish ground truth before any code is written. Run The Nautikal prompts manually. Know your Week 0 score.

ID	Task	Effort	Priority	Deadline
T-01	Draft 25-prompt library for The Nautikal 4 categories: High Intent (10), Cruise-Line Specific (8), Problem/Use-Case (4), Product-Driven (3). Use the Signal v1.0 prompt list as a starting base.	Medium	P0	Jun 18
T-02	Run all 25 prompts manually on ChatGPT + Gemini Copy-paste results into a spreadsheet. Record: brand mentioned? (Y/N), rank position, screenshot. This is your Week 0 baseline.	Small	P0	Jun 20
T-03	Build The Nautikal Truth Registry (JSON) Approved claims, disallowed claims, product specs (pricing, compatibility), policy facts (shipping, returns), canonical source URLs. Store in brand_config JSON.	Medium	P0	Jun 22
T-04	Set up Supabase project + run schema migrations Create tables: prompts, raw_runs, scores, brand_config. Free tier sufficient for MVP. Enable Row Level Security for future multi-tenancy.	Small	P0	Jun 24
T-05	Scaffold Python monorepo + API key setup aurenix/ monorepo. Folders: agents/prompt_runner, shared/, scheduler/, data/. Verify OpenAI + Gemini API keys. Test rate limits. Store keys in .env.	Small	P0	Jun 26

Phase 1 Jun 29 - Jul 10, 2026

Prompt Runner (Agent 1)

Automate what you did manually in Phase 0. Prompts run on a schedule, responses land in Supabase, zero manual steps.

ID	Task	Effort	Priority	Deadline
T-06	Build OpenAI + Gemini API adapters Unified adapter interface. Each adapter handles: auth, rate limiting (token bucket), exponential backoff (3 retries), response normalization to standard JSON.	Medium	P0	Jul 2
T-07	Build Prompt Runner pipeline (simple Python worker, no LangGraph yet) Load prompts from DB -> call adapters -> normalize response -> store in raw_runs. Skip LangGraph - use sequential Python with Cron trigger. Add run_batches health tracking.	Medium	P0	Jul 6
T-08	Set up Cron scheduler (Railway or local) Daily run at 6 AM UTC. Configurable via .env. Add a cost cap guard (\$10/day default). Run deduplication: skip if same prompt+model ran <6h ago.	Small	P1	Jul 8
T-09	Run health dashboard (Streamlit - minimal) Display: batch status, success rate %, cost/run estimate, last run timestamp, failure log. Not the GEO dashboard - just operational health.	Small	P1	Jul 10

DONE WHEN: 100 prompts/day running across ChatGPT + Gemini. >95% success rate. Results queryable in Supabase.

Phase 2 Jul 13 - Jul 31, 2026

Response Judge & GEO Scoring (Agent 2)

Turn raw AI responses into structured scores. Produce the first real GEO Score for The Nautikal.

ID	Task	Effort	Priority	Deadline
T-10	Build Inclusion scorer (fuzzy brand mention detection) Regex + fuzzy matching for: 'The Nautikal', 'Nautikal', 'thenautikal.com', product names. Score: 0 = absent, 50 = indirect, 100 = explicitly named.	Small	P0	Jul 16
T-11	Build Rank scorer Parse ordered/unordered lists. Detect brand position. Rule: if Inclusion = 0, Rank Score = 0. Prominence bonus: first mention in first paragraph = +10.	Small	P0	Jul 16
T-12	Build Accuracy checker (LLM-as-judge vs Truth Registry) Use Gemini Flash as judge. Prompt: classify each AI claim about The Nautikal as CORRECT / INCORRECT / UNSUPPORTED / HALLUCINATED. Output structured JSON issues.	Medium	P0	Jul 20
T-13	Build Citation + Sentiment scorers Citation: extract URLs, classify owned/earned/third-party. Sentiment: LLM-as-judge framing analysis (positive/neutral/mixed/negative) with key phrase evidence.	Medium	P1	Jul 23
T-14	Build composite GEO Score + Share-of-Voice rollups MVP formula: $(0.25 \times \text{Inclusion}) + (0.25 \times \text{Rank}) + (0.20 \times \text{Accuracy}) + (0.15 \times \text{Citation}) + (0.15 \times \text{Sentiment})$. SoV = brand prompts / total prompts x 100 by category.	Medium	P0	Jul 27
T-15	Manual QA: score first 25 runs, validate output Compare LLM-as-judge scores to human judgment on 25 sampled responses. Flag and fix any systematic errors. Calibrate scoring rubric before shipping dashboard.	Small	P0	Jul 31

DONE WHEN: Stable dashboard showing: Inclusion %, Avg Rank, Accuracy %, Share-of-Voice. Before/After comparison view works.

Phase 3 Aug 3 - Aug 14, 2026

GEO Dashboard (MVP UI)

Make the data visible and actionable. The dashboard is your demo asset and internal command center.

ID	Task	Effort	Priority	Deadline
T-16	Build GEO Dashboard Home (scores + trends) AI Visibility Score (composite), Inclusion Rate, Avg Rank, per-engine breakdown (ChatGPT vs Gemini). Weekly trend line. Insights panel: strong areas, gaps.	Medium	P0	Aug 7
T-17	Build Prompt Analysis table All 25 prompts. Columns: prompt, mentioned (Y/N), rank, GEO score, priority tag (HIGH/MED/LOW). Clickable rows open Viewer.	Small	P0	Aug 10
T-18	Build AI Response Viewer + Before/After comparison Show full AI response for any prompt. Side-by-side before/after when re-runs exist. Highlight brand mentions in-text.	Medium	P1	Aug 12
T-19	Build Recommendations tab (manual input for MVP) Manually populated list of 5-10 recommendations from scoring data. Not automated yet - just structured display with priority, rationale, target page, expected lift.	Small	P1	Aug 14

DONE WHEN: Working dashboard hosted internally. Can demo to a brand client end-to-end: here is your AI visibility score, here is what is failing, here is what to fix.

Phase 4 Aug 17 - Sep 4, 2026

GEO Recommender (Agent 3 MVP) + Full Loop

Close the loop. Generate prioritized recommendations automatically, implement one content change on The Nautikal, re-run, measure delta.

ID	Task	Effort	Priority	Deadline
T-20	Build Gap Detection engine (6 gap types) Missing Comparison Page, Missing Policy FAQ, Weak Differentiation, Inconsistent Naming, Missing Structured Data, Content Freshness. Auto-detect from scores + issues table.	Medium	P0	Aug 21
T-21	Build ICE Prioritization + Backlog generator Priority = Impact (1-5) x Confidence (1-5) x (6 - Effort) / 25 x 100. Output: ranked geo_backlog table. Link each item to the prompt_ids it targets.	Medium	P0	Aug 24
T-22	Build Execution Kit generator (AI-drafted briefs) Per backlog item: title, rationale, target URL, draft H1/H2 outline, FAQ blocks, JSON-LD schema template, expected score lift. Reuse geo-skill-for-agents schema templates.	Medium	P1	Aug 28
T-23	Close the loop: implement 1 fix on The Nautikal, re-run, measure delta Pick the highest-priority backlog item. Implement content change on thenautikal.com. Re-run affected prompts. Compute before/after GEO Score delta. Document case study.	Medium	P0	Sep 1
T-24	Weekly digest automation (Slack/Telegram) Automated weekly report: top 3 gains, top 3 gaps, #1 recommended action. Delivered via Slack webhook or Telegram bot.	Small	P2	Sep 4

DONE WHEN: Full loop proven: content change -> re-run -> measurable GEO Score improvement documented. The Nautikal case study is your first sales asset.

Signal v1.0 - Definition of Done

- 25 prompts running daily on ChatGPT + Gemini with >95% success rate
- GEO Score computed for The Nautikal: Inclusion %, Avg Rank, Accuracy %, Share-of-Voice
- Dashboard shows trend data with per-prompt breakdown
- At least 1 content change implemented on thenautikal.com with measurable score delta
- First AureniX case study documented: Before/After GEO Score comparison

Signal v2.0 - Full Commercial Product

The complete AI Discovery Optimization platform. Built on the validated v1.0 foundation, expanded to 4 AI engines, multi-brand multi-tenant, with social discovery intelligence, an experimentation engine, and a white-label agency tier.

v1.0	v2.0
x 25 prompts / 2 engines	✓ 100+ prompts / 4 engines (+ Perplexity + Claude)
x Simple Python workers	✓ LangGraph state machines with retry + human-in-loop
x Manual recommendations	✓ Automated Agent 3 with ICE-scored backlog
x Single brand (Nautikal)	✓ Multi-brand, multi-tenant, white-label
x Streamlit internal dashboard	✓ Full Next.js dashboard + REST API
x No social signals	✓ Reddit + YouTube signal extraction layer
x No experiments	✓ Full GEO Experimentation Engine with lift attribution
x Internal use only	✓ Subscription SaaS sold to brands + agencies

Product Modules

AI Visibility Engine (v2.0 Launch)

Continuous multi-engine monitoring

- 100+ prompts/day across ChatGPT, Gemini, Perplexity, and Claude
- 5-dimension GEO Scoring: Inclusion, Rank, Accuracy, Citation, Sentiment
- Composite GEO Score 0-100 with trend history
- Share-of-Voice vs named competitors
- Real-time alerts: lost citation positions, new hallucinations, competitor surges
- Regional + language variant testing (US, UK, India)

Truth Registry & Hallucination Guard (v2.0 Launch)

Brand accuracy enforcement layer

- Managed Truth Registry per brand (claims, specs, policies, pricing)
- LLM-as-judge fact verification on every AI response
- Hallucination classification: CORRECT / INCORRECT / UNSUPPORTED / HALLUCINATED
- Issues log with evidence quotes + source traceability
- Human QA sampling queue (5-10% of runs flagged weekly)
- Score stability alerts: >15-point composite swings trigger review

GEO Recommender & Backlog (v2.0 Launch)

From scores to prioritized actions

- Automated gap detection: 6 gap types (Comparison, Policy, Differentiation, Naming, Schema, Freshness)
- ICE prioritization: Impact x Confidence x (6 - Effort)
- Full execution kits: brief, H1/H2 outline, FAQ blocks, JSON-LD schema templates
- Social content prompts aligned to AI discovery queries
- Expected score lift estimates per backlog item
- Before/After tracking: re-run after changes, measure delta automatically

Command Center Dashboard (v2.0 Launch)

Brand AI presence at a glance

- GEO Score trend lines: daily, weekly, monthly views
- Per-prompt performance breakdown with intent tagging
- Competitor SoV comparison dashboard
- Before/After response viewer with in-text brand highlights
- Backlog tracker: recommended -> in progress -> verified lift
- Exportable weekly PDF reports (ReportLab gauges + priority actions)

Social Discovery Intelligence (v2.1 (Q1 2027))

The signals that move AI citations

- Reddit signal extraction: r/cruise, r/travel, r/royalcaribbean + brand-relevant subs
- YouTube summary mining: cruise packing videos, product mention frequency
- TikTok trend capture: captions, hashtags, emerging product vocabulary
- Trend velocity scoring: keyword frequency, momentum, community sentiment
- Social signal feed into GEO Analyzer: Social Popularity Score dimension
- Explains WHY AI recommends certain products (source attribution)

GEO Experimentation Engine (v2.1 (Q1 2027))

Test. Measure. Prove it works.

- A/B test content changes: track which versions move GEO scores
- Experiment types: structured product pages, FAQ sections, comparison pages, schema markup
- Control group isolation: hold-out prompt sets to detect organic drift vs. deliberate change
- Lift attribution: which content change drove which score improvement
- Knowledge graph gap experiments: entity disambiguation, brand claim standardization
- Experiment result library: what worked, what did not, by industry/category

Multi-Brand & Agency Platform (v2.2 (Q2 2027))

Built to scale from day one

- Multi-tenant architecture: per-brand prompt libraries, truth registries, dashboards
- White-label dashboard for agency clients
- Brand onboarding kit: self-serve Truth Registry builder, prompt library templates by vertical
- REST API: expose scoring data to client BI tools
- Role-based access: agency admin, brand manager, read-only stakeholder
- Billing per brand seat with shared infrastructure economics

Platform Intelligence (LangGraph Orchestration) (v2.2 (Q2 2027))

Full agent coordination layer

- LangGraph state machines replacing simple Python workers
- Human-in-the-loop approval nodes for content publishing
- Parallel subagent execution: 5 scoring dimensions run simultaneously
- Persistent run state: resume failed batches, no data loss
- Google AI Overview detection via Serper.dev integration
- Automated content publishing: approved FAQ blocks pushed to CMS via API

v2.0 Build Sequence (Post-MVP)

Sep-Oct 2026

- Expand to Perplexity + Claude API adapters
- Port Python workers to LangGraph state machines
- Build 100-prompt library (all 7 categories from Blueprint)
- Upgrade Streamlit -> Next.js + Recharts dashboard

Nov-Dec 2026

- Full Agent 3 automation (gap detection + execution kits)
- Competitor SoV dashboard
- PDF weekly reports (ReportLab pipeline from geo-skill-for-agents)
- v2.0 launch: onboard first paying client

Jan-Mar 2027 (v2.1)

- Reddit + YouTube social signal ingestion
- GEO Experimentation Engine with lift attribution
- Real-time alerts: lost citations, competitor surges, hallucination detection
- Google AI Overview tracking via Serper.dev

Apr-Jun 2027 (v2.2)

- Multi-tenant architecture + brand onboarding self-serve kit
- White-label dashboard + REST API
- Agency tier launch: 10-brand accounts
- Automated content publishing via CMS API integration